

JENKINS MASTER-SLAVE

Jenkins Master-Slave (Agent) architecture is a distributed setup where the Master manages jobs and the Slave (Agent) executes the assigned build tasks.

Agent Definition

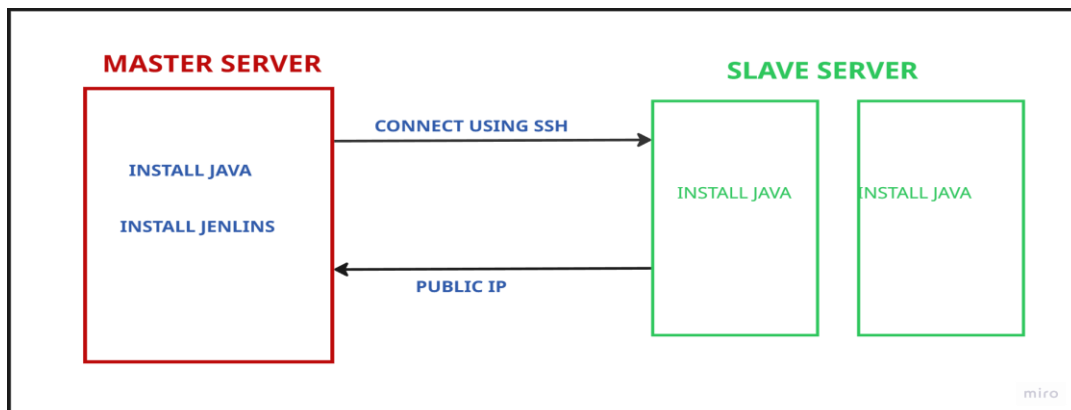
An Agent is a machine connected to the Jenkins Master that performs build, test, and deployment tasks assigned by the Master.

Label Definition

A Label is a tag assigned to an agent node that allows Jenkins to run specific jobs on a particular agent.

Benefits of Jenkins Master-Slave Architecture

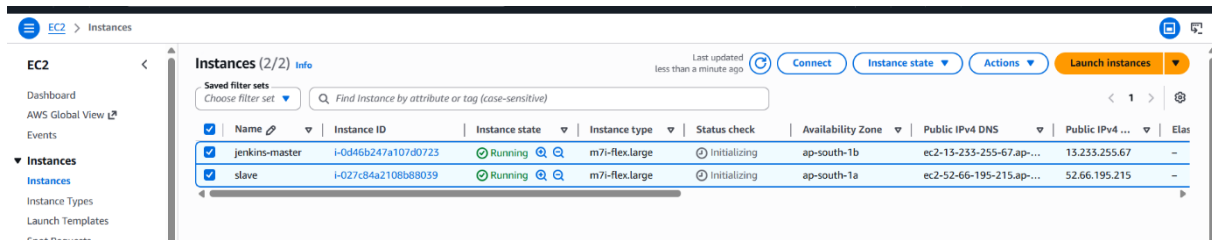
- Distributes build workload across multiple agents.
- Improves build performance and scalability.
- Allows different environments for different jobs.
- Reduces the load on the Jenkins Master.
- Enables parallel execution of multiple builds.



HANDS-ON FOR JENKINS MASTER - SLAVE

Screenshot: Create Master and Slave Ubuntu Instances

- Launch two Ubuntu EC2 instances in AWS.
- Name one instance as **Master** and the other as **Slave**.
- Both Ubuntu instances are created successfully and are in the running state.



Screenshot: Install and Start Jenkins on Master Instance

- Install **Java** and **Jenkins** on the Master instance.
- Start the Jenkins service using `sudo systemctl start jenkins`.
- Verify the service status using `sudo systemctl status jenkins`.

```
root@ip-172-31-2-102:/home/ubuntu# sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
39 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  fontconfig openjdk-21-jre

Installing dependencies:
  adwaita-icon-theme  gsettings-desktop-schemas  libavahi-common3  libglib2.0-0  libibus-1.0-5
  alsa-topology-conf  gtk-update-icon-cache  libcairo-gobject2  libglib2.0-dev  libice6
```

```
root@ip-172-31-2-102:/home/ubuntu#
root@ip-172-31-2-102:/home/ubuntu# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
--2026-06-02 03:50:07-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc 100%[=====
```

```
ubuntu@ip-172-31-14-76:~$ sudo su
root@ip-172-31-14-76:/home/ubuntu# systemctl start jenkins
root@ip-172-31-14-76:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-06-09 06:20:16 UTC; 6min ago
   Invocation: 2ae845475bfc4e738dc666c34045c7a1
   Main PID: 661 (java)
     Tasks: 43 (limit: 8157)
    Memory: 740.7M (peak: 741M)
       CPU: 24.211s
```

Screenshot: Install Java on Slave Instance

- Install **Java** on the Slave instance.
- Verify the installation using `java -version`.

```
applicable law.
```

```
ubuntu@ip-172-31-36-28:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

i-027c84a2108b88039 (slave)

PublicIPs: 52.66.195.215 PrivateIPs: 172.31.36.28

Screenshot: Generate SSH Key on Master Instance

- Run ssh-keygen on the Master instance.
- Press **Enter** for the file location.
- Press **Enter** for the passphrase.
- Press **Enter** again to confirm the passphrase.
- SSH key pair generated successfully.

```
root@ip-172-31-14-76:/home/ubuntu# exit
exit
ubuntu@ip-172-31-14-76:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:fmQO64t5DenPX+OjOH09B4LVu/bDsW+Z26BTCiBAJLA ubuntu@ip-172-31-14-76
The key's randomart image is:
+--[ED25519 256]--+
|...oo          |
| .  .  .       |
|E . . . o .    |
|   ...= . .    |
|   .Sooo .     |
|   .O.O . =.   |
|   .. o =+=    |
|   oooo.=o=O*  |
|   o...+oo=.o  |
+-----[SHA256]-----+
```

Screenshot: Verify SSH Key Files

- Run ls -la.
- Navigate to the .ssh directory using cd .ssh.
- Run ls.
- SSH key files (id_rsa and id_rsa.pub) are displayed successfully.

```
ubuntu@ip-172-31-14-76:~$ ls -la
total 36
drwxr-x--- 5 ubuntu ubuntu 4096 Jun  4 12:36 .
drwxr-xr-x 3 root   root   4096 Jun  4 11:10 ..
-rw----- 1 ubuntu ubuntu   8 Jun  4 12:36 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Feb 13 12:16 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 12:16 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Jun  4 11:12 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Feb 13 12:16 .profile
drwx----- 2 ubuntu ubuntu 4096 Jun  9 06:29 .ssh
drwxr-xr-x 5 root   root   4096 Jun  4 11:56 testenv
ubuntu@ip-172-31-14-76:~$ cd .ssh
ubuntu@ip-172-31-14-76:~/.ssh$ ls
authorized_keys id_ed25519 id_ed25519.pub
```

Screenshot: Display and Copy Public SSH Key

- Run `cat id_ed25519.pub`.
- Copy the displayed public SSH key.

```
ubuntu@ip-172-31-14-76:~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIB07Zj+Vx3Lv6+IFyWT6EWzN2TOKlCRM3bLYOZGNCpAt ubuntu@ip-172-31-14-76
ubuntu@ip-172-31-14-76:~/.ssh$
```

Screenshot: Configure Public Key on Slave

- Run `cd .ssh`
- Run `ls`
- Run `nano authorized_keys`

```
ubuntu@ip-172-31-36-28:~$ cd .ssh
ubuntu@ip-172-31-36-28:~/.ssh$ ls
authorized_keys
ubuntu@ip-172-31-36-28:~/.ssh$ nano authorized_keys
ubuntu@ip-172-31-36-28:~/.ssh$
```

Screenshot: Paste Public Key in authorized_keys

- `nano authorized_keys`
- Paste the copied public key.
- Press **Ctrl + X**.
- Press **Y** to save the changes.
- Press **Enter** to exit the editor successfully.

```
GNU nano 8.7.1 authorized_keys *
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCA2co/bHicdEoH1FbWQC+e1Q0wB/bURp15uTupCX3u3z3wE2z11nyITKwicWueKYhHN430/WO2ND3S60YVFH1085v08OmYrOWJ8FzuFid1PtJD12HIC05IG6g1BRzRq/9nUCFCrn2KgY9z15S22
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIB07Zj+Vx3Lv6+IFyWT6EWzN2TOKlCRM3bLYOZGNCpAt ubuntu@ip-172-31-14-76
```

```
⌕ Help      ⌕ Write Out  ⌕ Where Is   ⌕ Cut        ⌕ Execute   ⌕ Location  ⌕ Undo      ⌕ Set Mark  ⌕ To Bracket ⌕ Previous  ⌕ Back
⌕ Exit      ⌕ Read File  ⌕ Replace    ⌕ Paste      ⌕ Justify   ⌕ Go To Line ⌕ Redo      ⌕ Copy      ⌕ Where Was  ⌕ Next      ⌕ Forward

i-027c84a2108b88039 (slave)
PublicIPs: 52.66.195.215 PrivateIPs: 172.31.36.28
```

Screenshot: Copy Slave Public IP

- Select the **Slave** instance.
- Copy the **Public IP Address**.

```
i-027c84a2108b88039 (slave)
PublicIPs: 52.66.195.215 PrivateIPs: 172.31.36.28
```

Screenshot: Connect Master to Slave via SSH

- Run `ssh ubuntu@<Slave_Public_IP>`

- Type yes and press **Enter**.
- Master connected to the Slave instance successfully

```
ubuntu@ip-172-31-14-76:~/.ssh$
ubuntu@ip-172-31-14-76:~/.ssh$ cd ..
ubuntu@ip-172-31-14-76:~$ ssh ubuntu@52.66.195.215
```

i-0d46b247a107d0723 (jenkins-master)

PublicIPs: 13.233.255.67 PrivateIPs: 172.31.14.76

.....

```
ubuntu@ip-172-31-14-76:~/.ssh$
ubuntu@ip-172-31-14-76:~/.ssh$ cd ..
ubuntu@ip-172-31-14-76:~$ ssh ubuntu@52.66.195.215
The authenticity of host '52.66.195.215 (52.66.195.215)' can't be established.
ED25519 key fingerprint is: SHA256:r2TFzQY8SY841UcyD1keAg6dQ4es40HKxhKODGBqofk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.66.195.215' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)
```

```
* Documentation: https://docs.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

System information as of Tue Jun 9 06:40:31 UTC 2026

```
System load: 0.0      Temperature:      -273.1 C
Usage of /:  40.5% of 6.61GB Processes:         158
Memory usage: 4%      Users logged in:   0
Swap usage:  0%      IPv4 address for enp39s0: 172.31.36.28
```

Expanded Security Maintenance for Applications is not enabled.

```
51 updates can be applied immediately.
46 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
Last login: Tue Jun 9 06:23:13 2026 from 13.233.177.4
ubuntu@ip-172-31-36-28:~$
```

Screenshot: Verify SSH Connection

- Verify the **Last login** message displays ubuntu@<Slave_Private_IP>.
- Verify that the displayed IP matches the **Slave Private IP**
- SSH connection verified successfully.

```
Last login: Tue Jun 9 06:23:13 2026 from 13.233.177.4
ubuntu@ip-172-31-36-28:~$
```

i-027c84a2108b88039 (slave)

PublicIPs: 52.66.195.215 PrivateIPs: **172.31.36.28**

Screenshot: Create Jenkins Directory on Slave

- Run `mkdir jenkins`
- Run `cd jenkins`

- Run `pwd`
- Copy the displayed path successfully.

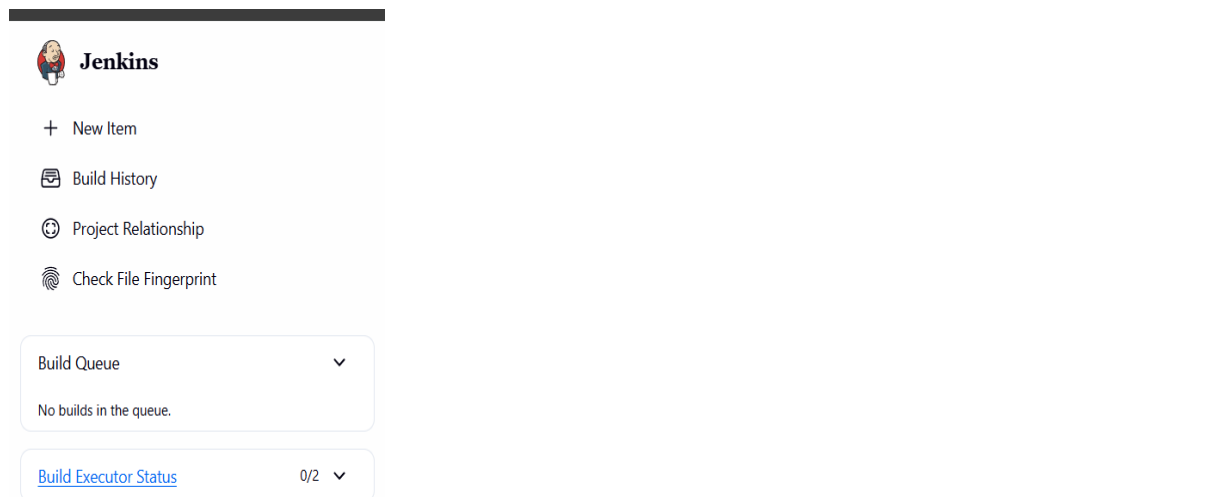
```
ubuntu@ip-172-31-36-28:~/.ssh$ cd ..
ubuntu@ip-172-31-36-28:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-36-28:~$ mkdir jenkins
ubuntu@ip-172-31-36-28:~$ cd jenkins/
ubuntu@ip-172-31-36-28:~/jenkins$ pwd
/home/ubuntu/jenkins
ubuntu@ip-172-31-36-28:~/jenkins$
```

i-027c84a2108b88039 (slave)

PublicIPs: 52.66.195.215 PrivateIPs: 172.31.36.28

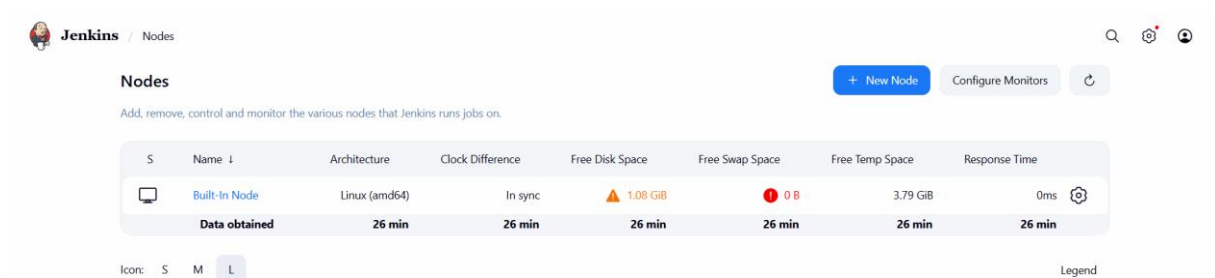
Screenshot: Open Build Executor Status

- Open the **Jenkins Dashboard**.
- Click **Build Executor Status**.



Screenshot: Add a New Jenkins Node

- Click the **+ New Node** button.



Screenshot: Create a New Agent Node

- Enter the node name as **agent**.
- Select **Permanent Agent**.
- Click **Create**.

New node

Node name

agent

Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

```
ubuntu@ip-172-31-36-28:~/jenkins$ pwd
/home/ubuntu/jenkins
ubuntu@ip-172-31-36-28:~/jenkins$
```

i-027c84a2108b88039 (slave)

Screenshot: Configure Agent Node

- Set **Number of executors** to **2**.
- Paste the copied **Remote root directory** path.
- Set **Labels** as **new**.
- Select **Use this node as much as possible** under **Usage**.

Number of executors ?

2

Remote root directory ?

/home/ubuntu/jenkins

Labels ?

new

Usage ?

Use this node as much as possible

Use this node as much as possible

Only build jobs with label expressions matching this node

Screenshot: Configure Launch Method

- Select **Launch agents via SSH** as the **Launch method**.
- Paste the **Slave Public IP Address** in the **Host** field.

Launch method ?

Launch agents via SSH

Host ?

52.66.195.215

Screenshot: Add SSH Credentials

- Click **+ Add** under **Credentials**.
- Select **Jenkins**.
- Select **Global credentials (unrestricted)**.

Credentials ?

- none - + Add

! The selected credentials cannot be found

Host Key Verification Strategy ?

System

Global
Credentials that should be available everywhere.

Screenshot: Select SSH Credentials Type

- Select **SSH Username with private key**.
- Click next

Add Credentials ×

Select a type of credential

 Username with password
Commonly used for authentication to services like Git, APIs, or registries.

 GitHub App

 SSH Username with private key

 Secret file

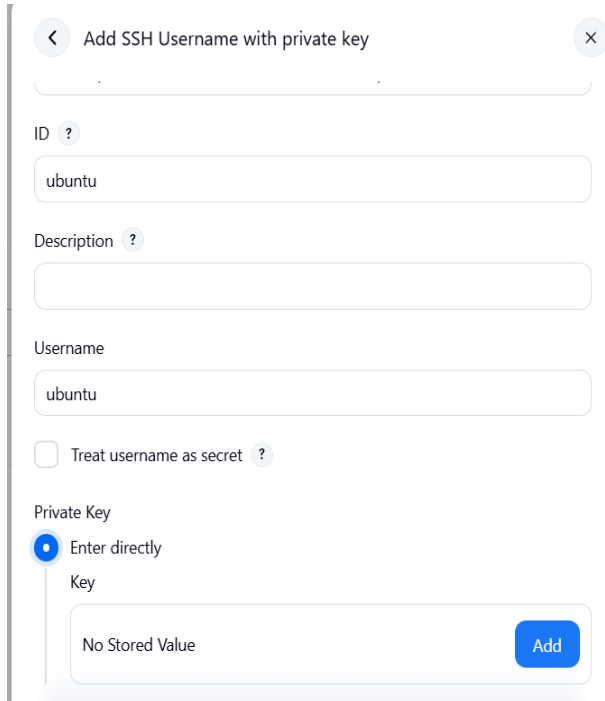
 Secret text

 Certificate
Upload a PKCS#12 or PEM encoded certificate and private key.

Next

Screenshot: Configure SSH Credentials

- Enter **ID** as ubuntu.
- Enter **Username** as ubuntu.
- Under **Private Key**, click **Add**.



< Add SSH Username with private key X

ID ?
ubuntu

Description ?

Username
ubuntu

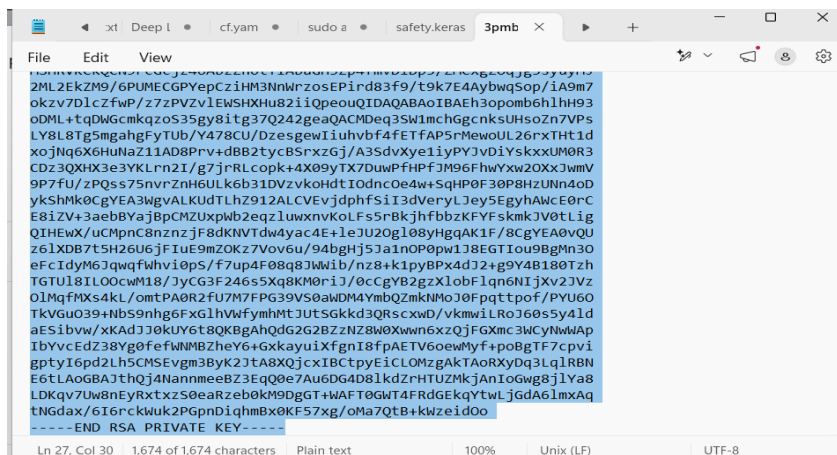
☐ Treat username as secret ?

Private Key
☒ Enter directly

Key
No Stored Value **Add**

Screenshot: Add Private Key

- Open the **.pem private key** file in Notepad.
- Copy the private key.



```
-----BEGIN RSA PRIVATE KEY-----
2ML2EKZM9/6PUMECGPYepCziHM3NnWrzosEP1rd83f9/t9k7E4AybwqSop/1A9m7
okzv7D1cZfwP/z7zPVZv1EWSHXHu821iQpeouQIDAQABaoIBAEh3opomb6h1hH93
oDML+tqDwGcmkqzoS35gy81tg37Q242geaQACMDeq3SW1mchGgcnskUHS0zn7VPS
LY8L8Tg5mgahgFyTub/Y478CU/Dzesgew1iuhvbf4FETfAP5rMewoUL26rxTht1d
xoJNq6HuNaZ11AD8Prv+dB82tycBSrxzGj/A3SdvXye1iyPYJvDiYskxxUM0R3
CDz3QHIX3e3YKLrn2I/g7jrRLcopk+4X09yTX7DuwPfHPfJM96FhwYxw20XxJwmV
9P7Fu/zPQss75nvrZnH6ULk6b31DVzvkoHdtIodncOe4w+SqHP0F30P8HzUNn4oD
ykShMk0cGEA3WgvALKudTLhZ912ALCvEvjdpHfS1I3dVeryLJey5EgyhAwcE0rC
E8iZV+3aebBYaJBpCMZUxpwb2eqz1uwXnvKofS5rBkjhfbbzKFYfSkmkJV0tLlg
QIHEwX/ucMpnC8nzjF8dKNVTdw4yac4E+1eJU20g108yHgqAK1F/8CgYEA0vQU
z61XDB7t5H26U6jfIuE9mZOKz7Vov6u/94bgHj5Ja1nOP0pw1J8EGTIou9BgMn3O
eFcIdym6Jqwfwhvi0ps/f7up4F08q8JwIb/nz8+k1pyBPx4dJ2+g9Y4B180Tzh
TGTU18ILO0cwM18/JycG3F246s5Xq8KM0riJ/0cCgYB2gzXlobf1qn6NIjXv2JvZ
O1MqfMXs4kl/omtPA0R2fu7M7FP639V50aWDM4YmbQZmkNM0J0Fpqttptof/PYU6O
TKVGU039+Nb59nhg6FXglhVwfymhMTJUTSGkkd3QRscxwD/vkmmw1LROJ60s5y41d
aESibvw/xKAdJJ0kUY6t8QKBgAhQdG2G2BzNZ8w0Xwn6xzQJFGXmc3WkYNwAp
IbYvcEdZ38Yg0fefWNB8ZheY6+GxkayuiXfgnI8fPAETV6oewMyf+po8gTf7cpvi
gptyI6pd2Lh5CMSEvgm3ByK2JtA8XQjcxIBCTpyEiCLOmZgAKTAORxyDq3Lq1RBN
E6tLa0BAJthQj4NannmeeBZ3Eq0e7Au6DG4D81kdZrHTUZMkjAnIoGwg8j1Ya8
LDKqv7Uw8nEyrxtxzS0earZeb0kM9DgGT+wAFT0GT4FRdGEkqYtLwJgDA61mxAQ
tNGdax/616rckwuk2PGpnDihmBx0KF57xg/oMa7QtB+kwzeid0o
-----END RSA PRIVATE KEY-----
```

Ln 27, Col 30 | 1,674 of 1,674 characters | Plain text | 100% | Unix (LF) | UTF-8

Screenshot: Create SSH Credentials

- Paste the copied private key into the **Private Key** field.
- Click **Create**.

Private Key

☒ Enter directly

Key

Enter New Secret Below

```

9P7tU/7PQSS75hVrZnH6ULK6b31DVZVKOHat10ancOe4w+SqHP0F30P
8HzUNn4oD
ykShMk0CgYEA3WgvALKudTLhZ912ALCvEvjdphfSiI3dVeryLJey5Eg
yhAWcE0rC
E8iZV+3aebBYajBpCMZUxpWb2eqzluwxnvKoLFs5rBkjhfbbzKFYFsk
mkJV0tLig
QIHEwX/uCMpnC8nznzjF8dKNVTdw4yac4E+leJU20gl08yHgqAK1F/8
CgYEA0vQU
z6lXDB7t5H26U6jFIuE9mZOKz7Vov6u/94bgHj5Ja1nOP0pw1J8EGTI
ou9BgMn3O
eFcIdyM6Jqwqfwhvi0pS/f7up4F08q8JWwib/nz8+k1pyBPx4dJ2+g9

```

Create

Screenshot: Select SSH Credentials

- Select **ubuntu** from the **Credentials** dropdown.

Credentials ?

ubuntu

jaisurya07/*****

ubuntu

Screenshot: Configure Agent Connection

- In **Host Key Verification Strategy**, select **Manually trusted key**.
- In **Availability**, select **Keep this agent online as much as possible**.
- Click **Save**.

Host Key Verification Strategy ?

Manually trusted key Verification Strategy

☐ Require manual verification of initial connection ?

Advanced ▾

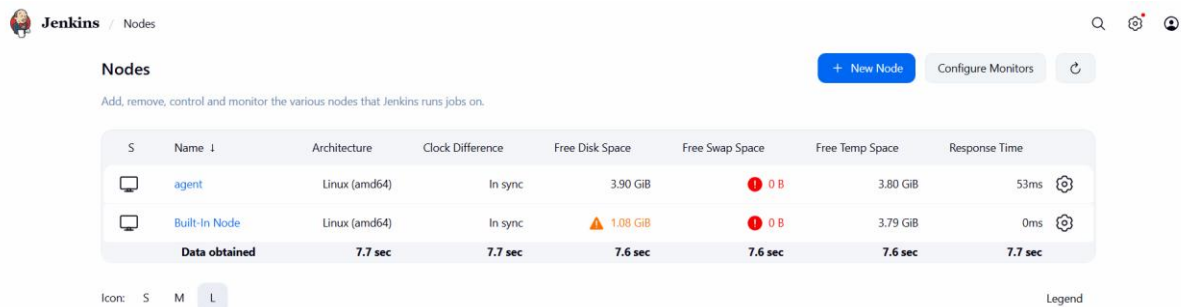
Availability ?

Keep this agent online as much as possible

Save

Screenshot: New Agent Node Created

- Agent node created successfully.
- Slave node is connected and ready to execute Jenkins jobs.

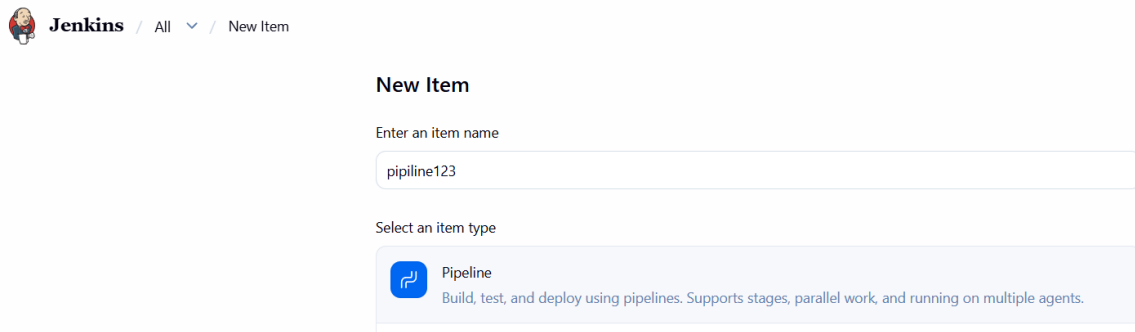


The screenshot shows the Jenkins 'Nodes' page. At the top, there's a breadcrumb 'Jenkins / Nodes' and a search icon. Below the header, there's a '+ New Node' button and a 'Configure Monitors' button. A sub-header 'Nodes' is followed by a description: 'Add, remove, control and monitor the various nodes that Jenkins runs jobs on.' Below this is a table with columns: S, Name, Architecture, Clock Difference, Free Disk Space, Free Swap Space, Free Temp Space, and Response Time. The table lists two nodes: 'agent' and 'Built-In Node'. The 'agent' node is in sync, has 3.90 GiB free disk space, 0 B free swap space, 3.80 GiB free temp space, and a response time of 53ms. The 'Built-In Node' is also in sync, has 1.08 GiB free disk space, 0 B free swap space, 3.79 GiB free temp space, and a response time of 0ms. Below the table, there's a 'Data obtained' row showing 7.7 sec for clock difference, 7.6 sec for free disk space, 7.6 sec for free swap space, and 7.7 sec for free temp space. At the bottom, there's a legend with icons for S, M, and L.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	agent	Linux (amd64)	In sync	3.90 GiB	0 B	3.80 GiB	53ms
	Built-In Node	Linux (amd64)	In sync	1.08 GiB	0 B	3.79 GiB	0ms
Data obtained			7.7 sec	7.7 sec	7.6 sec	7.6 sec	7.7 sec

Screenshot: Create a New Pipeline

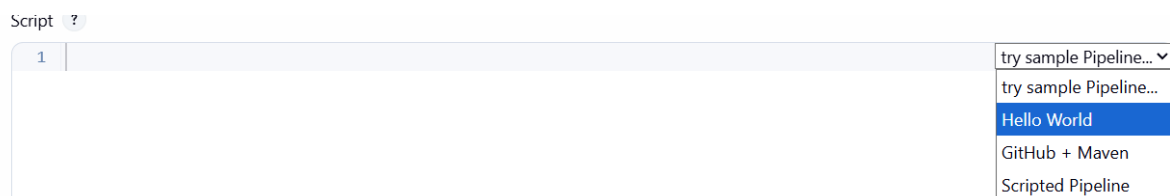
- Click **New Item**.
- Enter the project name.
- Select **Pipeline**.
- Click **OK**.
- Pipeline created successfully.



The screenshot shows the Jenkins 'New Item' page. At the top, there's a breadcrumb 'Jenkins / All / New Item'. Below the header, there's a 'New Item' section. Under 'Enter an item name', there's a text input field with 'pipeline123'. Under 'Select an item type', there's a dropdown menu with 'Pipeline' selected. The 'Pipeline' option is described as 'Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.'

Screenshot: Select Sample Pipeline

- In the Pipeline Script section, click Try Sample Pipeline.
- Select Hello World.



The screenshot shows the Jenkins 'Pipeline Script' section. At the top, there's a 'Script' label with a help icon. Below it, there's a text area for the script. A dropdown menu is open, showing options: 'try sample Pipeline...', 'try sample Pipeline...', 'Hello World', 'GitHub + Maven', and 'Scripted Pipeline'. The 'Hello World' option is selected.

Screenshot: Modify Pipeline Agent

- Delete agent any from the script.

- Add the following:

```
agent {  
    label "new"  
}
```

- Click Apply.
- Click Save.
- Pipeline updated successfully.

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent{  
3     label "new"  
4   }  
5  
6   stages {  
7     stage('Hello') {  
8       steps {  
9         echo 'Hello World'  
10      }  
11    }  
12  }  
13 }  
14
```

Save

Apply

Screenshot: Build Pipeline

- Click **Build Now**.
- Pipeline execution started successfully on the **agent node**.



Jenkins / pipeline123

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

📁 Stages

✎️ Rename

🔍 Pipeline Syntax

Builds >

🔍 Filter

Today

✅ #2 7:04 AM

Screenshot: Verify Build Execution on Agent Node

- Select the executed **Build**.
- Click **Console Output**.
- Verify that the build is running in the **remote root directory** configured for the newly created **agent node**.
- Build execution on the agent node verified successfully.

 **Jenkins** / pipeline123 ▾ / #2 ▾

Status

</> Changes

 Console Output

 Edit Build Information

 Delete build '#2'

 Timings

 Pipeline Overview

 Restart from Stage

 Replay

 Pipeline Steps

 Workspaces

 Previous Build

Console

```
Started by user jai
[Pipeline] Start of Pipeline
[Pipeline] node
Running on agent in /home/ubuntu/jenkins/workspace/pipeline123
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello World
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

JENKINS USER MANAGEMENT

Jenkins User Management is the process of creating and managing user accounts, assigning roles, and controlling access permissions to ensure secure access to the Jenkins server.

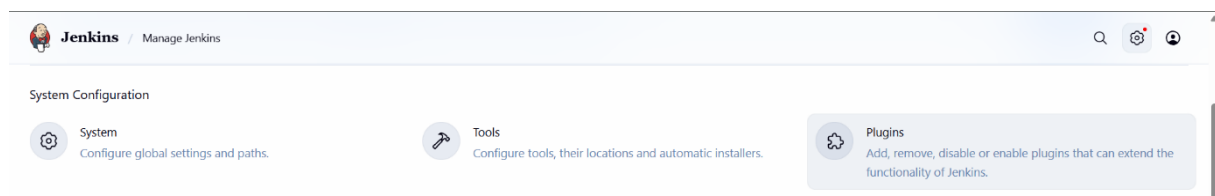
Uses

- Controls user access to Jenkins resources.
- Enhances security by assigning appropriate permissions to different users.

HANDS-ON FOR JENKINS USER MANAGEMENT

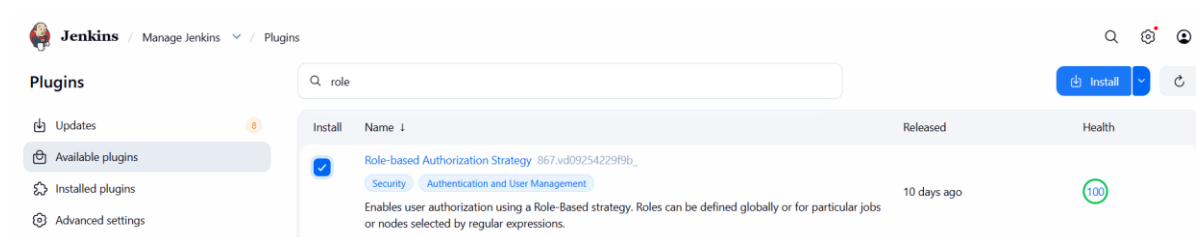
Screenshot: Open Jenkins Plugins

- Open the **Jenkins Dashboard**.
- Click the **Settings** icon.
- Select **Plugins**.



Screenshot: Install Role-Based Authorization Plugin

- Go to the **Available Plugins** tab.
- Search for **Role-based Authorization Strategy**.
- Select the plugin.
- Click **Install**.
- Plugin installed successfully.



.....

Download progress

Preparation

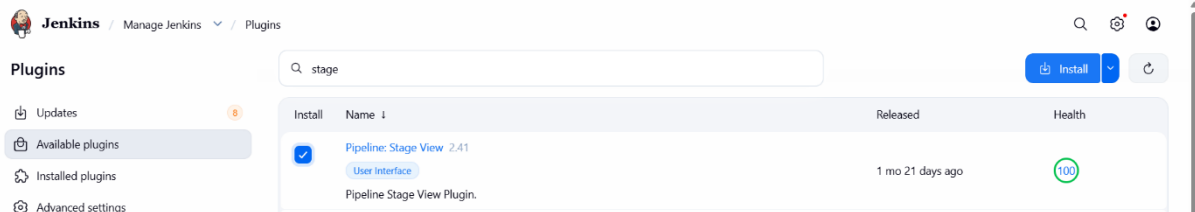
- Checking internet connectivity
- Checking update center connectivity
- Success

Role-based Authorization Strategy  Success

Loading plugin extensions  Success

Screenshot: Install Pipeline Stage View Plugin

- Search for **Pipeline Stage View** in **Available Plugins**.
- Select the plugin.
- Click **Install**.
- Plugin installed successfully.

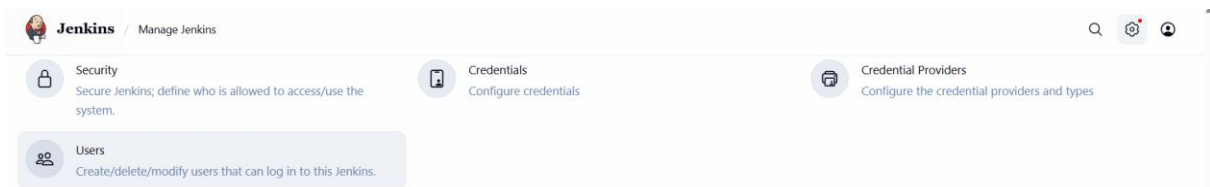


Download progress

Preparation	• Checking internet connectivity • Checking update center connectivity • Success
Role-based Authorization Strategy	✓ Success
Loading plugin extensions	✓ Success
Pipeline Graph Analysis	✓ Success
Pipeline: REST API	✓ Success
Pipeline: Stage View	✓ Success
Loading plugin extensions	✓ Success

Screenshot: Open Users Page

- Go to **Manage Jenkins**.
- Click **Users**.



Screenshot: Click Create User

- On the **Users** page, click **+ Create User**.
- Create User page opened successfully.



Screenshot: User Details

- Enter the **Username** as test
- Enter the **Password**.
- Enter the **Confirm Password**.
- Enter the **Full Name**.
- Enter the **Email Address**.
- User details entered successfully.

Create User

Username
test

Password
....

Confirm password
....

Full name
test

E-mail address
test@gmail.com

Create User

Screenshot: User Created Successfully

- Click **Create User**.
- Test User created successfully.

Users 2

+ Create User

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID	Name	
jai	jai	⚙️
test	test	⚙️ 🗑️

Screenshot: Create Another User

- Click **+ Create User**.
- Enter the user details for **dev**.
- Click **Create User**.

 **Jenkins** / Jenkins' own user database 🔍 ⚙️ 👤

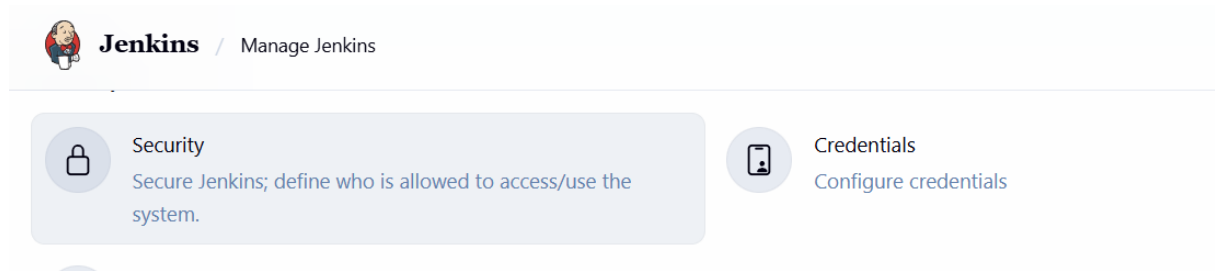
Users 3 + Create User

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID	Name	
dev	dev	⚙️ 🗑️
jai	jai	⚙️
test	test	⚙️ 🗑️

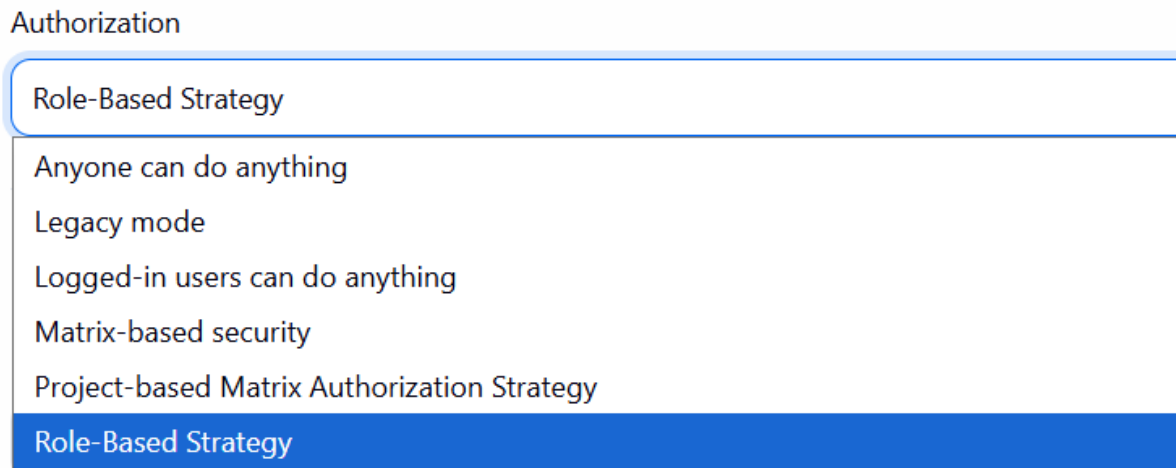
Screenshot: Open Security Settings

- Open **Manage Jenkins**.
- Click **Security**.



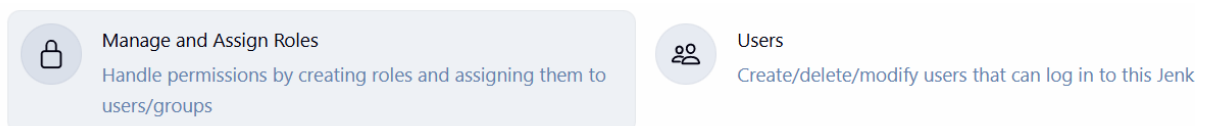
Screenshot: Select Role-Based Strategy

- Scroll down to the **Authorization** section.
- Select **Role-Based Strategy**.



Screenshot: Open Manage and Assign Roles

- Open **Manage Jenkins**.
- Click **Manage and Assign Roles**.



Screenshot: Add Dev Role

- In the **Role to add** field, enter **dev-role**.
- Click **Add**.
- Dev-role added successfully.

Jenkins / Manage Jenkins / Manage and Assign Roles / Manage Roles

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Role	Administrator	Manage	Read	Create	Delete	Update	inagDomains	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Cancel	Build	Configure	Cancel	Build	Discover	Move	Read	Workspace	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck	ThreadDump	View	Metrics
admin	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Role to add

dev-role

Add

Jenkins / Manage Jenkins / Manage and Assign Roles / Manage Roles

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Global roles

Role	Administrator	Overall	Credentials	Agent	Job	Run	View	SCM	Metrics
admin	☒	☐	☐	☐	☐	☐	☐	☐	☐
dev-role	☐	☐	☐	☒	☐	☒	☐	☒	☒

Role to add

dev-role

Screenshot: Add Test Role

- In the **Role to add** field, enter **test-role**.
- Click **Add**.
- Test-role added successfully.
- Select the required permissions **for test-role and dev-role**
- Permissions configured successfully for both roles.

Jenkins / Manage Jenkins / Manage and Assign Roles / Manage Roles

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Global roles

Role	Administrator	Overall	Credentials	Agent	Job	Run	View	SCM	Metrics
admin	☒	☐	☐	☐	☐	☐	☐	☐	☐
dev-role	☐	☒	☐	☒	☐	☒	☐	☒	☒
test-role	☐	☐	☐	☒	☐	☒	☐	☐	☐

Screenshot: Assign User to Dev Role

- Click **Assign Roles** next to **Manage Roles**.
- Click **Add User**.
- Enter the user id **dev**.

- Click **OK**.
- User added successfully.

Jenkins / Manage Jenkins / Manage and Assign Roles / Assign Roles

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Assign Roles

Global roles

User/Group	admin	dev-role	test-role
Anonymous	☐	☐	☐
Authenticated Users	☐	☐	☐
jai	☒	☐	☐

Add User Add Group

User ID:

Cancel OK

Screenshot: Add Test User

- Click **Add User**.
- Enter the user name **test**.
- Click **OK**.
- Test user added successfully.
- Assign **dev-role** to the user **dev**.
- Assign **test-role** to the user **test**.
- Roles assigned successfully to the users.

Assign Roles

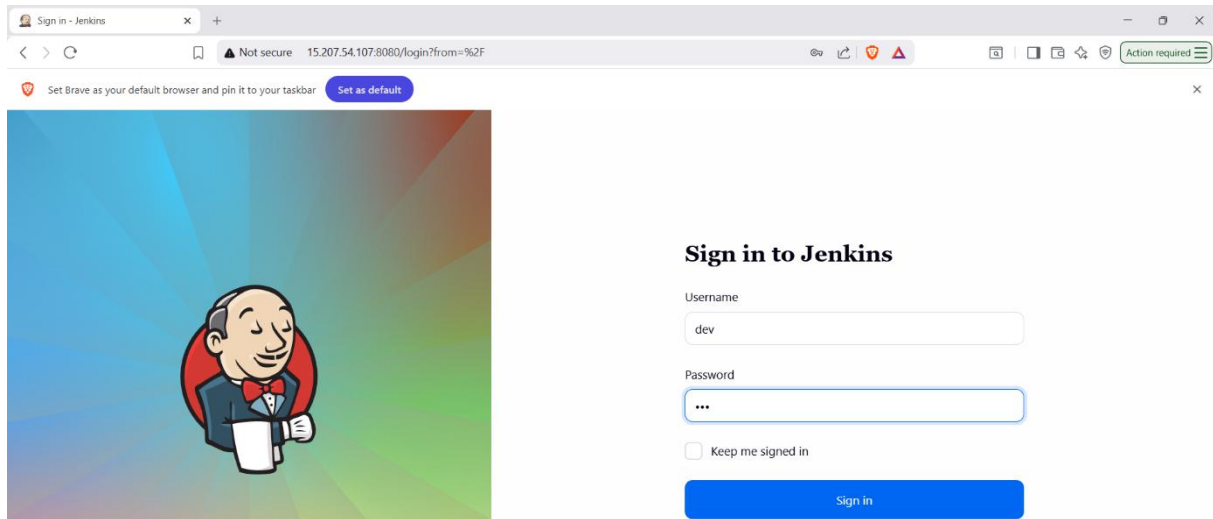
Global roles

User/Group	admin	dev-role	test-role
Anonymous	☐	☐	☐
Authenticated Users	☐	☐	☐
jai	☒	☐	☐
dev	☐	☒	☐
test	☐	☐	☒

Add User Add Group

Screenshot: Login as Dev User

- Open the Jenkins Dashboard in a new browser window.
- Enter the username **dev**.
- Enter the password **dev**.
- Click **Sign In**.



- Logged in successfully as the dev user.
- **dev** user with limited permissions.

